

Jaswanth KS

ML/AI Engineer

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PERSONAL PROFILE

Software Engineer with **3+** years of experience building **backend services**, **data pipelines**, and cloud-based applications using **Python** and **TypeScript**. Experienced in designing data ingestion systems, **REST APIs**, and data enrichment workflows across structured datasets. Provisioned and managed **AWS** infrastructure using **Terraform**, and worked with containerized deployments using **Docker**, with exposure to **Kubernetes**. Recently worked on prototype workflow using **LLM APIs** and retrieval-based approaches to explore document search, summarization, and enrichment across structured and document datasets. Completed an **MSc in Data Science and AI** (Merit), with a focus on **machine learning**, information retrieval, and **Applied AI** engineering. Interested in knowledge graphs, **Gen AI** systems, and scalable data infrastructure for financial data and analytics platforms.

EXPERIENCE

Software Engineer – Growz Software Solutions

Bengaluru, India

Nov 2021 – Aug 2023

- Designed and built **data ingestion** and **enrichment pipelines** integrating structured business and financial datasets into **AWS**-based data platforms, improving data accessibility and query performance by **30%**.
- Developed **microservices** and **REST APIs** using **Python** and **TypeScript**, to integrate external APIs and **ML** services, and automated data tagging workflows.
- Contributed to ML pipeline tooling and **RAG**-based workflows using **LangChain** and **LLM APIs**, supporting model evaluation and retrieval quality for internal datasets; collaborated with data science teams on output assessment and feature experimentation.
- Built monitoring dashboards to track **ETL pipeline** accuracy, latency, and job failures, improving pipeline reliability and incident detection by **50%**.
- Implemented **CI/CD** pipelines using **Git** and **Jenkins** and automated AWS infrastructure provisioning using **Terraform**, reducing deployment cycle time by **40%**.
- Collaborated with product and data teams to improve data modelling and support downstream analytics and ML workflows.

Tech Stack: Python, TypeScript, AWS, SQLAlchemy, Docker, Terraform, LangChain, GitHub, Pandas, NumPy, Scikit-learn, CI/CD, Pytest.

Software Engineer Intern – Techborn IT Solutions

Bengaluru, India

Sep 2020 – Oct 2021

- Developed **backend services** and **REST APIs** using **Python** to support predictive modelling workflows, analytics dashboards, and operational reporting tools.
- Assisted in developing and testing classification models using **TensorFlow**, and **Keras**. Built data preprocessing pipelines using **pandas** and **NumPy** for dataset organisation tasks.

Tech Stack: Python, Pandas, NumPy, Scikit-learn, TensorFlow, Keras, SQL Server, AWS (EC2, S3), REST APIs.

SKILLS

- **GenAI & Data Systems:** LLM API Integration, RAG concepts, Knowledge graphs and ontology
- **Data Engineering:** ETL Pipeline Development, Data Modelling, Query optimisation
- **Backend Engineering:** Python, TypeScript, JavaScript, FastAPI, Flask, Django, REST API

- **Cloud & DevOps:** AWS (Lambda, Glue, DynamoDB, RDS, S3, EC2), Terraform, Kubernetes, Docker, Jenkins, CI/CD, GitHub Actions
- **Databases:** PostgreSQL, SQL Server, MySQL, Data Modeling, Query Optimization
- **Testing:** Pytest, TDD, API Testing, Performance Monitoring

PROJECTS

NYSE Time Series Forecasting Model

- Built machine learning models using **ARIMA**, **SARIMA**, And **LSTM** to analyse and forecast stock price movements.
- Processed historical market data using **pandas** and evaluated predictions against real market data.
- Built interactive interface using **Flask** and **Plotly** to visualise predictions and Trends.

Tech Stack: Python, Pandas, NumPy, Statsmodels, TensorFlow/Keras, Flask, Plotly.

Building a National Lottery Profit/Loss Simulation Platform

- Built a full-stack simulation platform supporting multi-country lottery modelling and large-scale simulation runs.
- Implemented backend logic for authentication, simulation workflows, and statistical visualisation.
- Optimised system to support **80,000+** simulations with stable performance.

Tech Stack: Python, Flask, SQLAlchemy, SQLite, Plotly, Bootstrap, JavaScript.

Fruit Classification System using Transfer Learning

- Built a **CNN**-based image classification system using transfer learning with pre-trained **VGG16** and **ResNet50** architectures to classify fruit categories in real time.
- Applied data augmentation, **fine-tuning**, and feature extraction techniques to improve generalisation and achieve **93%** classification accuracy.
- Evaluated and compared model performance across architectures, analysing trade-offs between accuracy, inference speed, and computational cost.

Tech Stack: Python, TensorFlow/Keras, VGG16, ResNet50, OpenCV, NumPy.

EDUCATION

MSc in Data Science and Artificial Intelligence, Merit

University of Liverpool, United Kingdom

Sep 2023 – Jul 2025

- **Key Areas:** Machine Learning & Applied AI, Mathematics & Statistics for AI, Database & Information systems, Data Modelling
- Worked extensively with Python for Machine Learning, data analysis, and Applied AI workflows.

Bachelor of Science, Distinction

Sri Venkateswara University, India

Jun 2018 – Sep 2021

- **Key Areas:** Mathematics, Computer Science and Physics

CERTIFICATIONS

- Essentials in Github Professional Certificate - **GitHub**
- Programming with Python Professional Certificate - **OpenEDG Python Institute**
- Docker Foundations Professional Certificate - **Docker**
- Prompt Engineering for Generative AI – **LinkedIn**

LEADERSHIP & ACTIVITIES

- Led a team of 5 to secure a **top 3 finish** in a 24-hour hackathon, showcasing innovation and teamwork.
- Presented a technical paper at **SIGMOID-2K19**, effectively communicating complex concepts to a technical audience.